ROTARY ENCODER

Measured-data analysis

Motor identification campaign · 13 October 2026

A reader for students: evidence, calibration limits, and interpretation

WHAT WAS RECORDED

1,903,061 samples over 37.1 min, 83 actions, 20,000 counts/rev, 0 sequence gaps.

RUN INTEGRITY

Final residual +267 counts (+4.81°). Largest boundary change at action 59: 20 fs/s -.

STATIC MAP

Direction-pair mean spans 49.5 counts (0.89°); fitted 1/rev amplitude 21.2 counts (0.382°).

ECCENTRICITY EVIDENCE

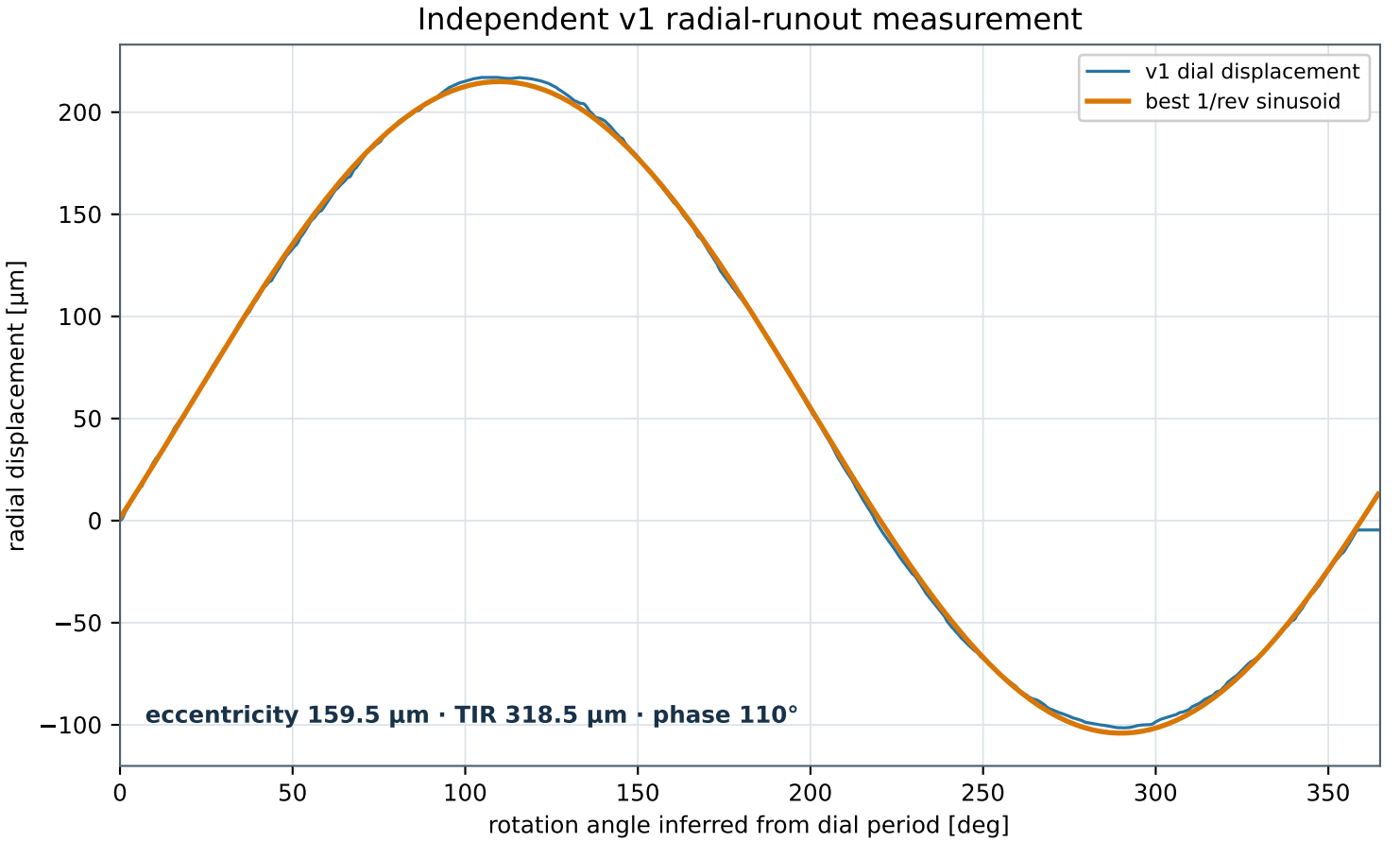
V1 dial result: 159.5 µm eccentricity, 318.5 µm TIR. It is not applied as angle calibration.

Central lesson. Encoder resolution is not absolute accuracy. Quantisation, radial runout, torque harmonics, friction, driver state, and lost position can occupy the same error trace. This report separates observations from inferences.

Reproduce with scripts/analyze_motor_encoder.py

What is measured—and what can be calibrated

The calibration decision is part of the result, not hidden preprocessing.



V1 RADIAL RESULT: NOT NUMERICALLY APPLIED TO ENCODER ANGLE

Centre offset e can create angular error proportional to e/R, where R is effective scale radius. R was not recorded, the high spot was not registered to encoder zero, and the instruments had no common clock. Applying 159.5 µm would manufacture an unknown amplitude and phase.

WHAT IS USED

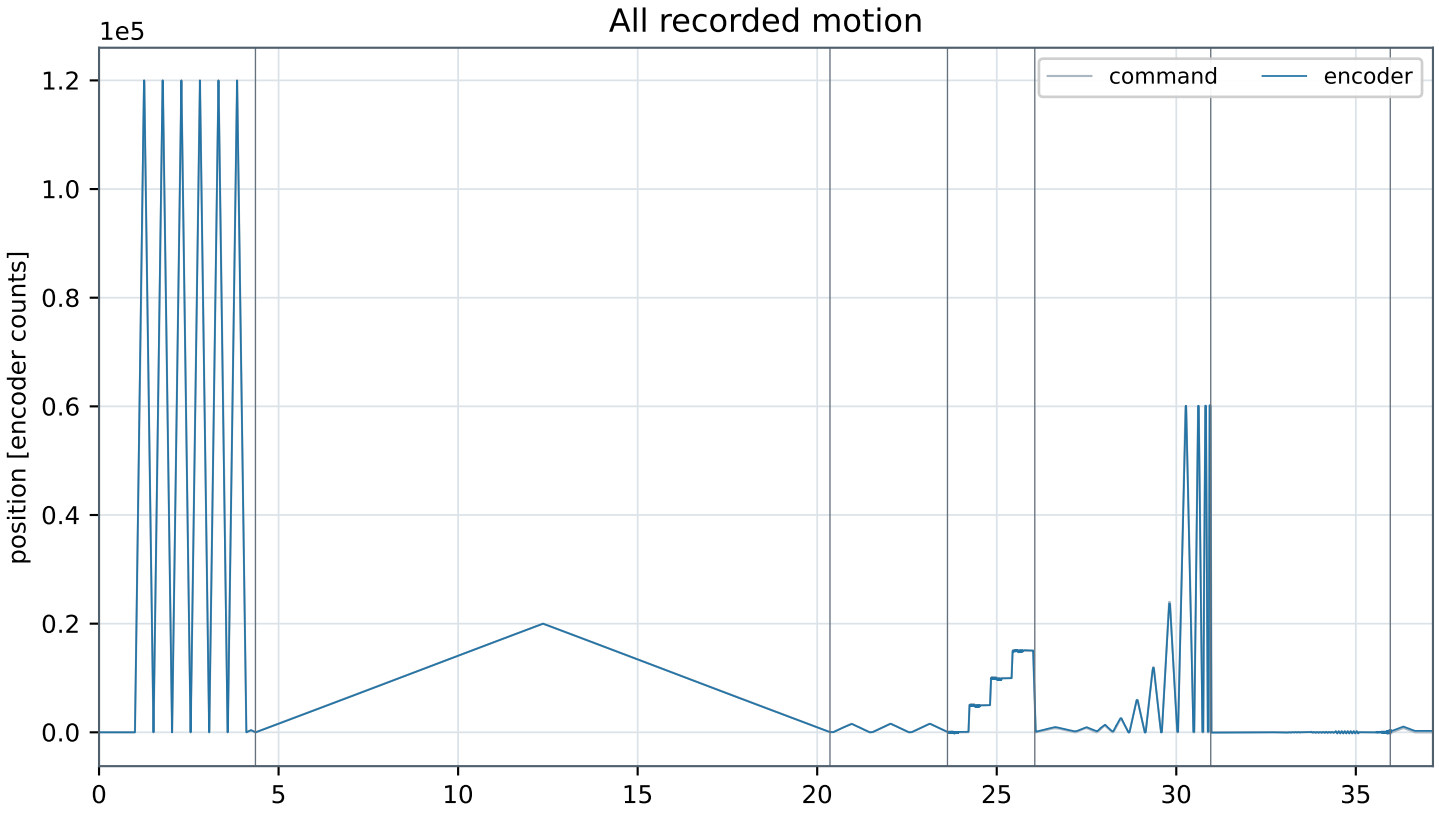
Raw encoder counts, commanded phase, direction pairs, action boundaries, and harmonic decomposition of the measured static map.

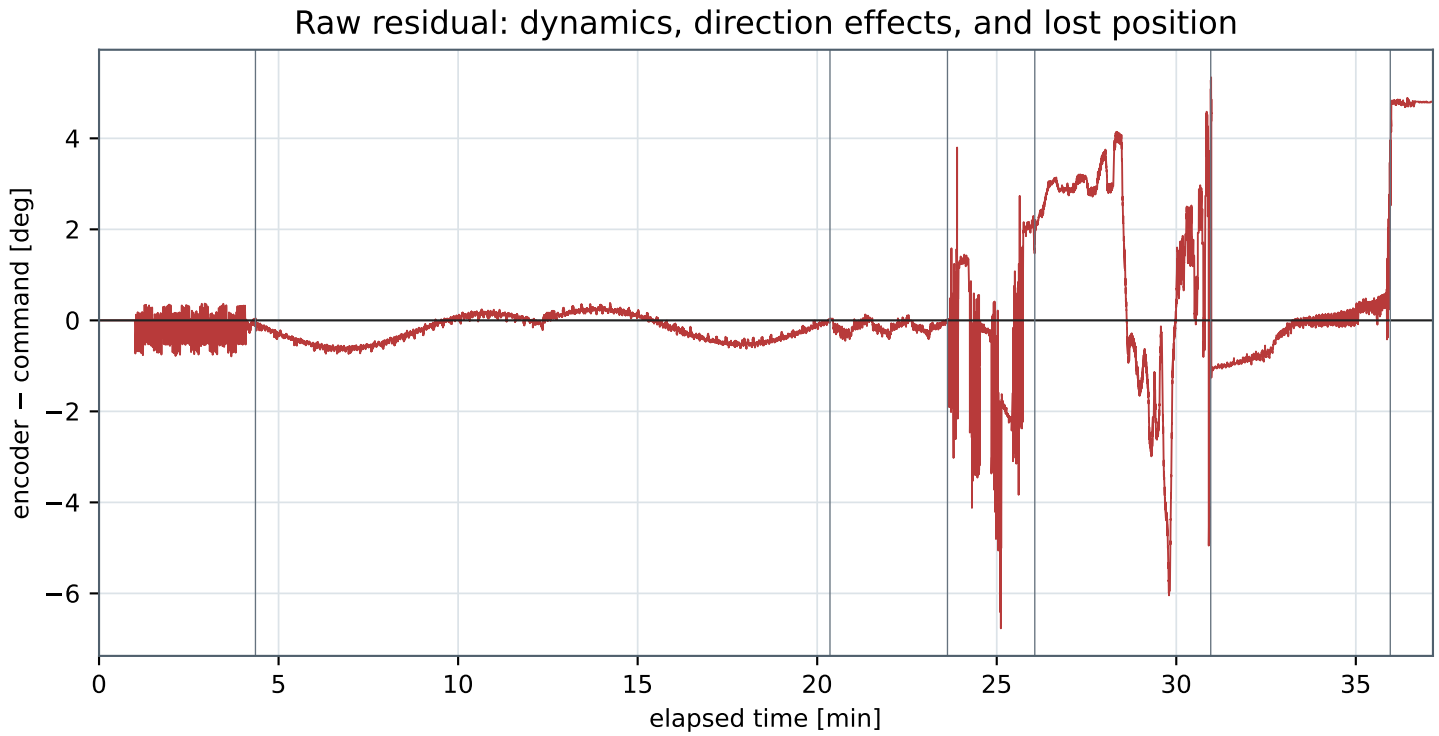
SENSITIVITY VIEW

Later pages subtract a fitted 1/rev term from the same static map, explicitly labeled diagnostic—not v1 calibration or ground truth.

Complete campaign overview

Command and encoder use the same 20,000-count-per-revolution scale.

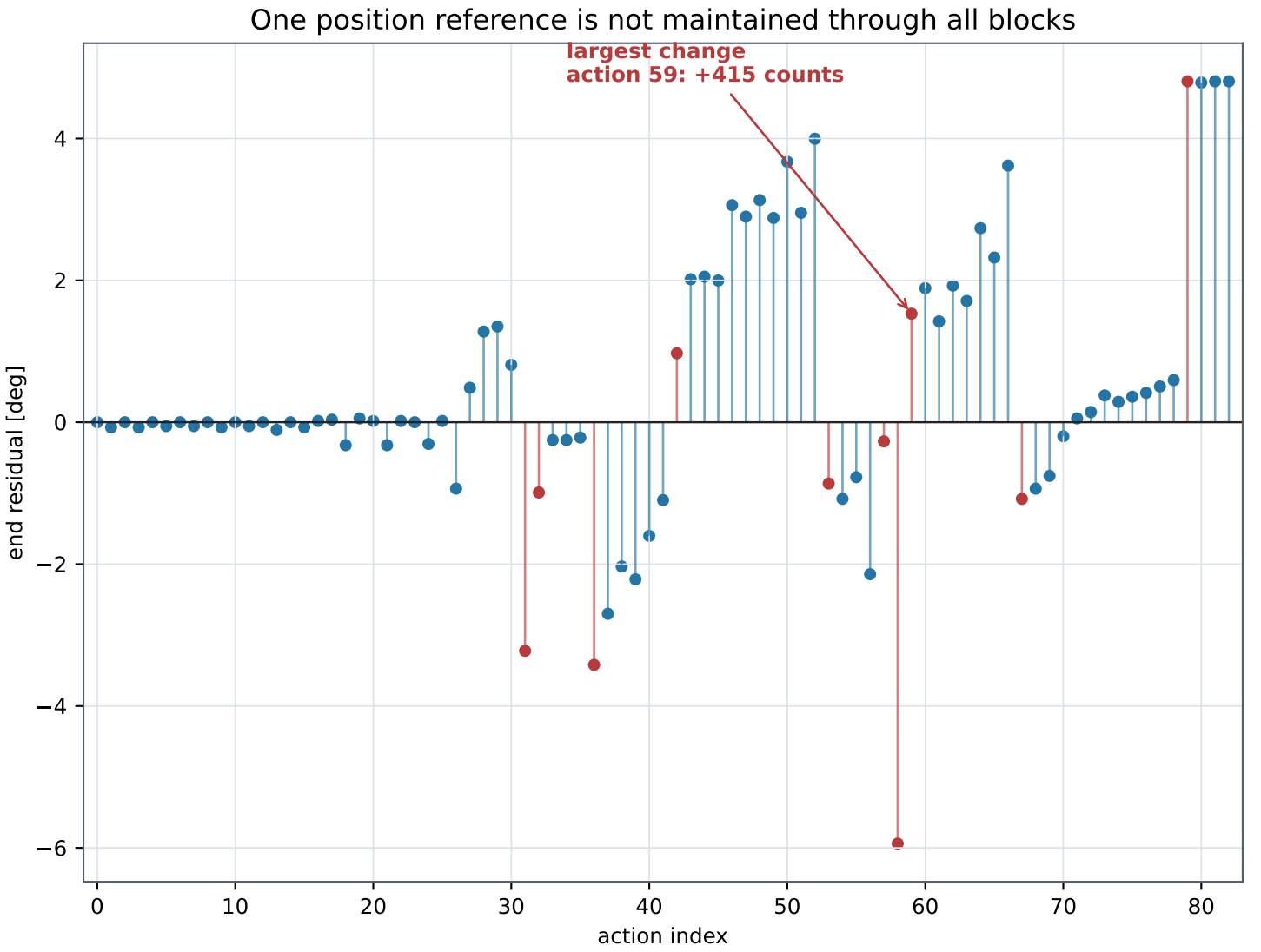




Excursions during motion are expected lag and settling. Persistent changes after an action ends instead indicate a changed reference or lost position. The next page separates these cases.

Position closure at every action boundary

Boundary residual = encoder minus command after the programmed action finishes.



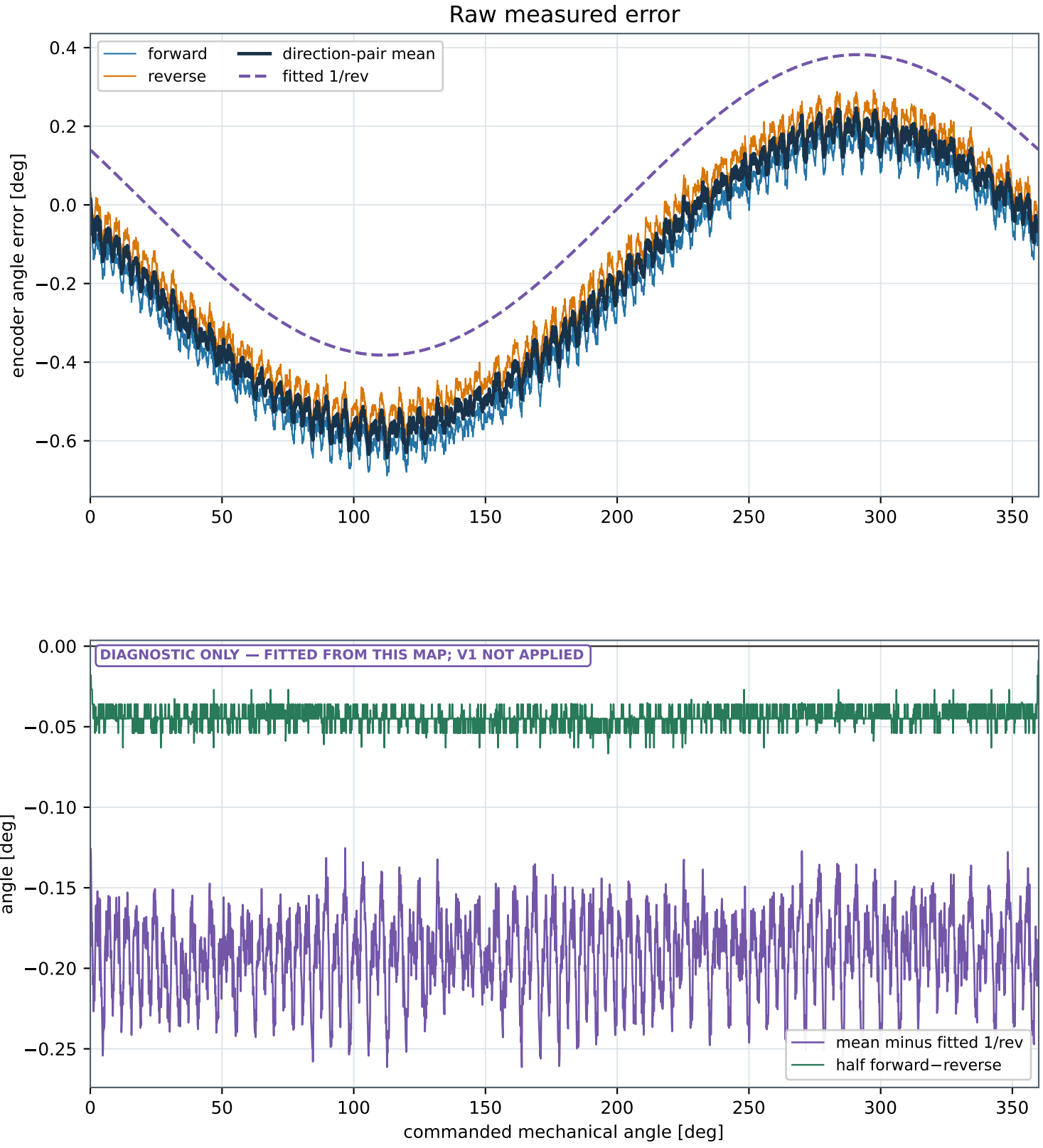
KEY INTEGRITY FINDING

Final residual +267 counts = +2.67 full steps = +4.81°. Largest change: action 59 (20 fs/s -), +415 counts. Later blocks require local interpretation.

The static map (actions 15–16) closes to 1 count and supports direction-paired analysis. The closing reference inherits offsets accumulated during later tests.

Quasi-static angle map: raw and sensitivity-corrected

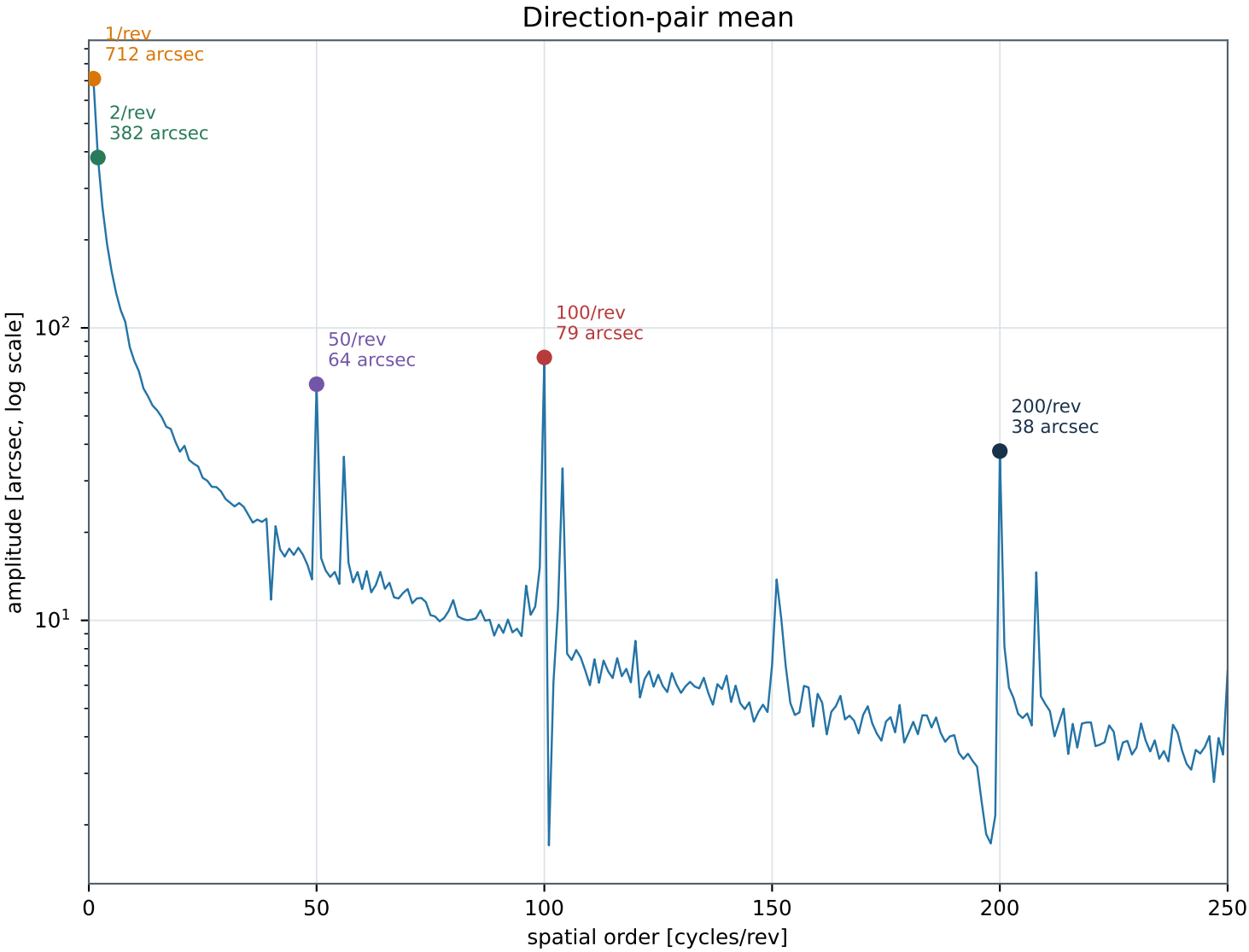
One shaft turn at native 1/16 microsteps; settled tail of every 150 ms dwell.



Raw mean span 0.890°. Fitted 1/rev amplitude 0.382° at phase 291.5°. Subtraction leaves 0.136° peak-to-peak, but also removes any real 1/rev motor error.

Spatial harmonic content of the static map

Cycles per mechanical revolution after removing offset and linear closure trend.

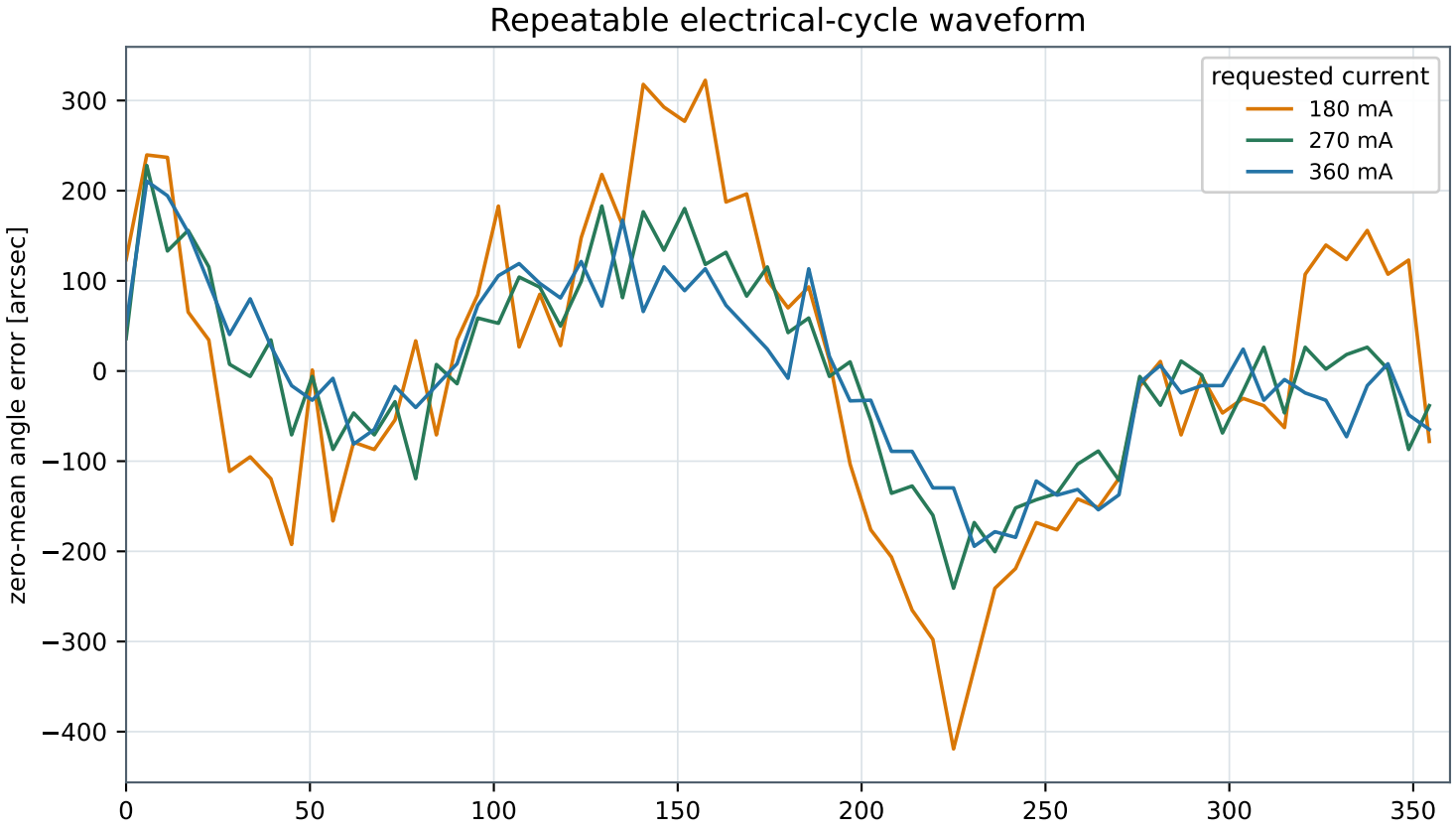


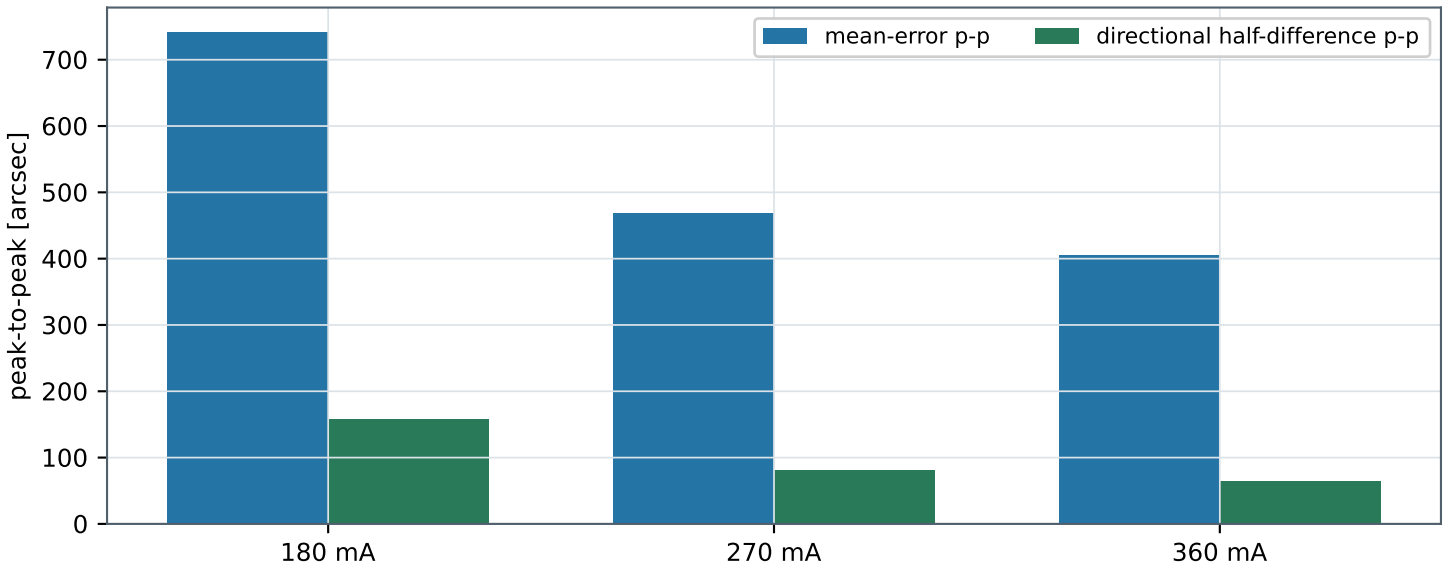
HOW TO READ THIS

Order 1 is consistent with eccentric mounting but not unique to it. Orders 50 and 100 correspond to 50 electrical cycles/rev and its second harmonic; order 200 coincides with full-step positions and detent/step structure. Periodicity does not prove mechanism.

Current dependence over the electrical cycle

Four cycles folded and averaged; forward/reverse means suppress directional friction.

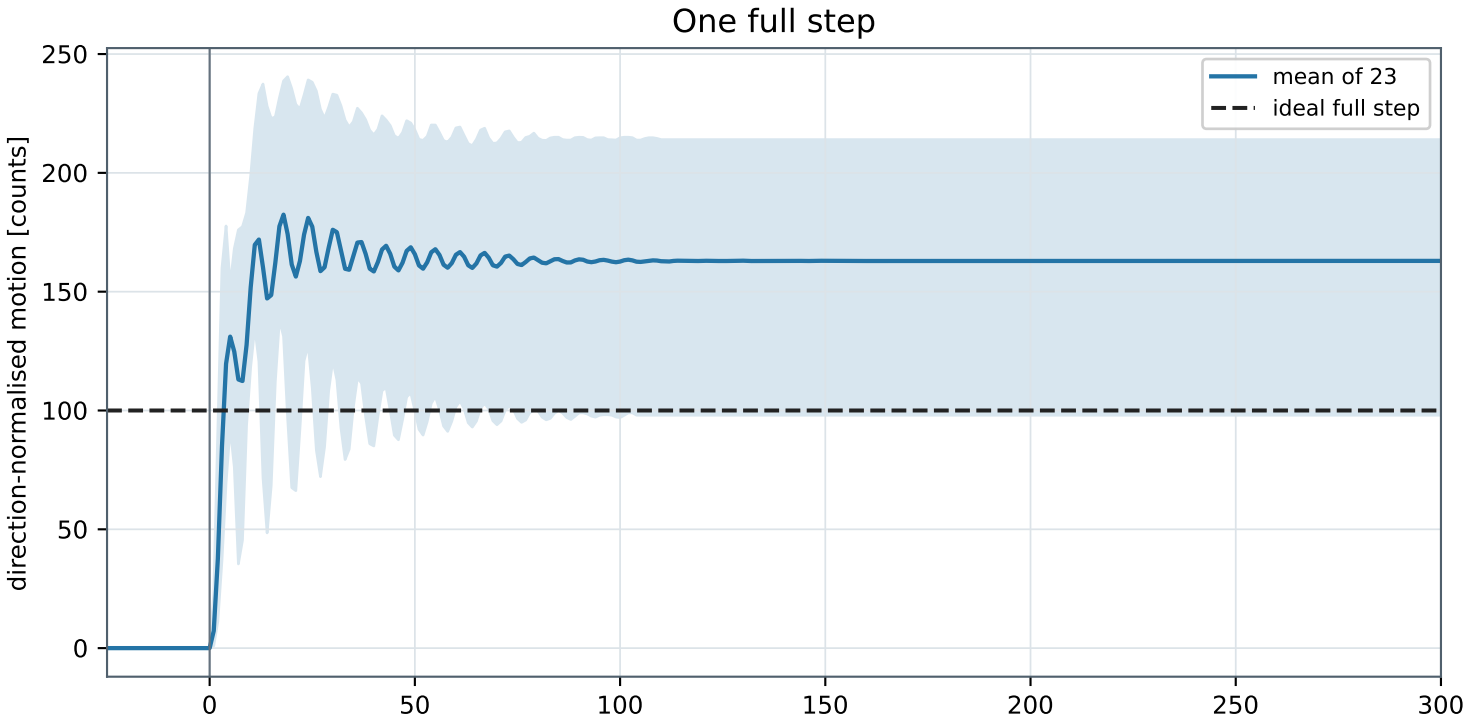


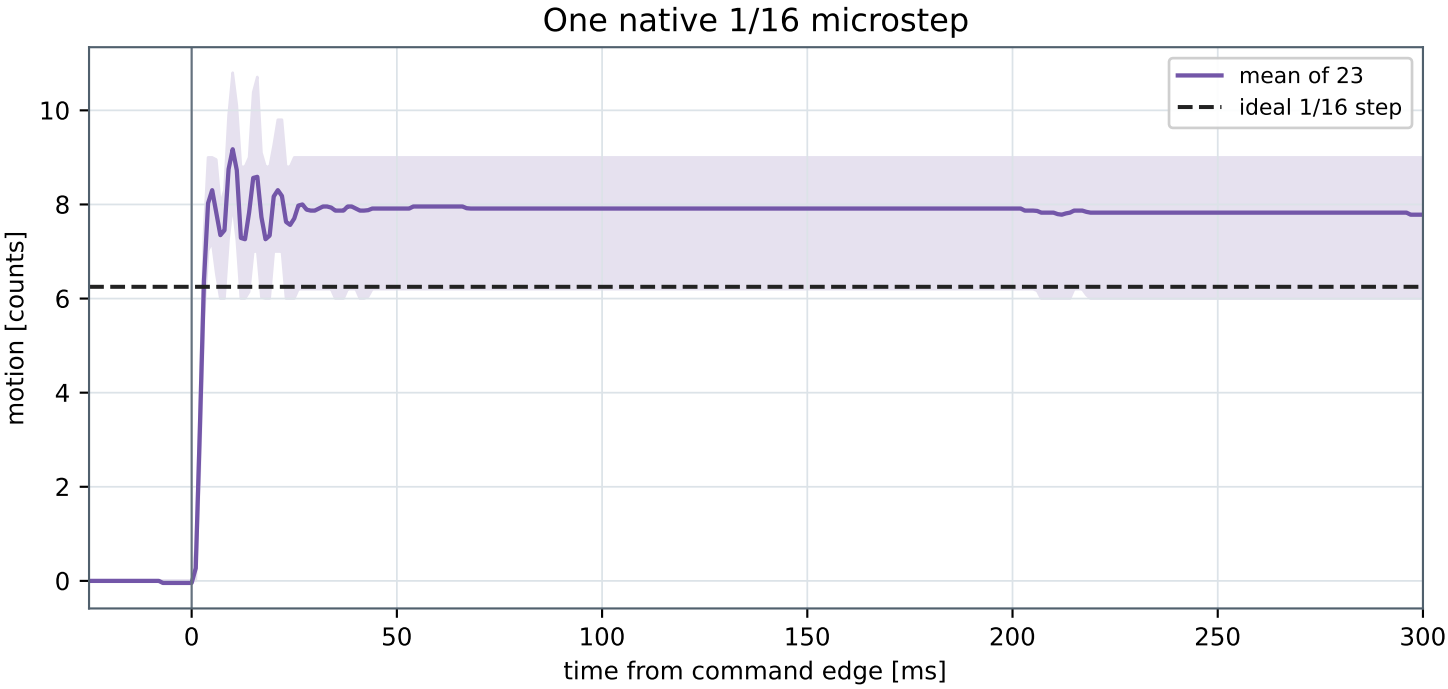


Detent-torque angle error is expected to shrink roughly as current rises; phase-current waveform distortion is more nearly current-independent. This sweep compares trends but does not identify either cause uniquely.

Step response at the zero-position test point

Repeated excitation edges aligned at t = 0; shading is the central 80%.

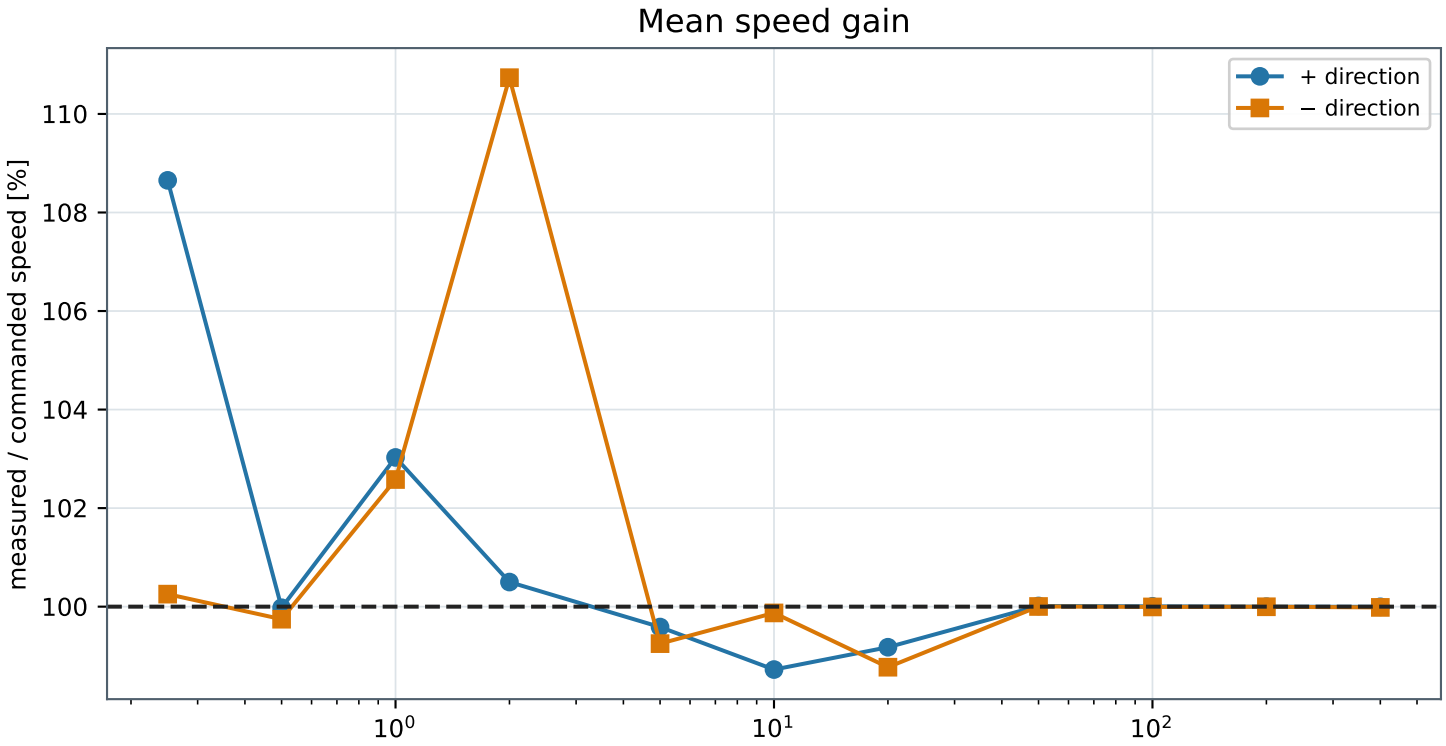


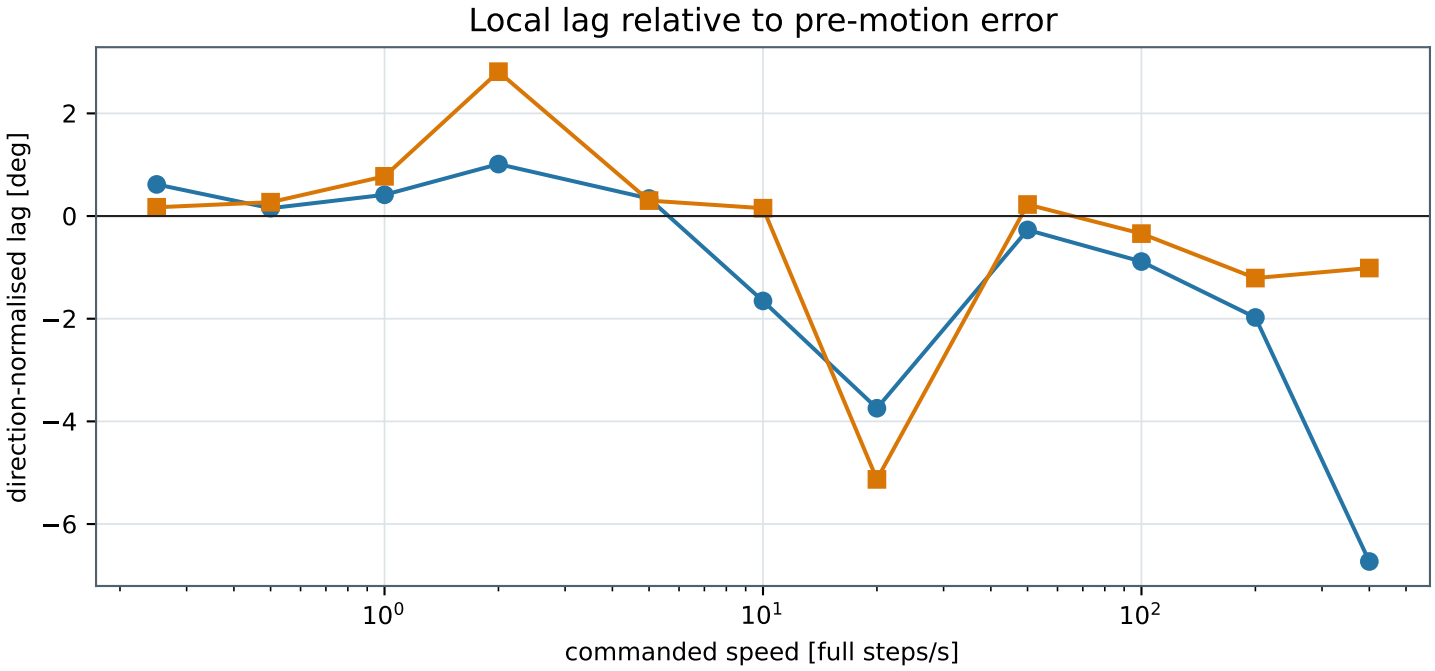


One encoder count is 0.018°. The ensemble settles near 163 counts for the nominal 100-count full step and 7.8 counts for the nominal 6.25-count microstep. These include the driver-state transition and should not be treated as a simple linear transfer function.

Constant-rate tracking

Linear fits use the central 80% between first and last step edges.

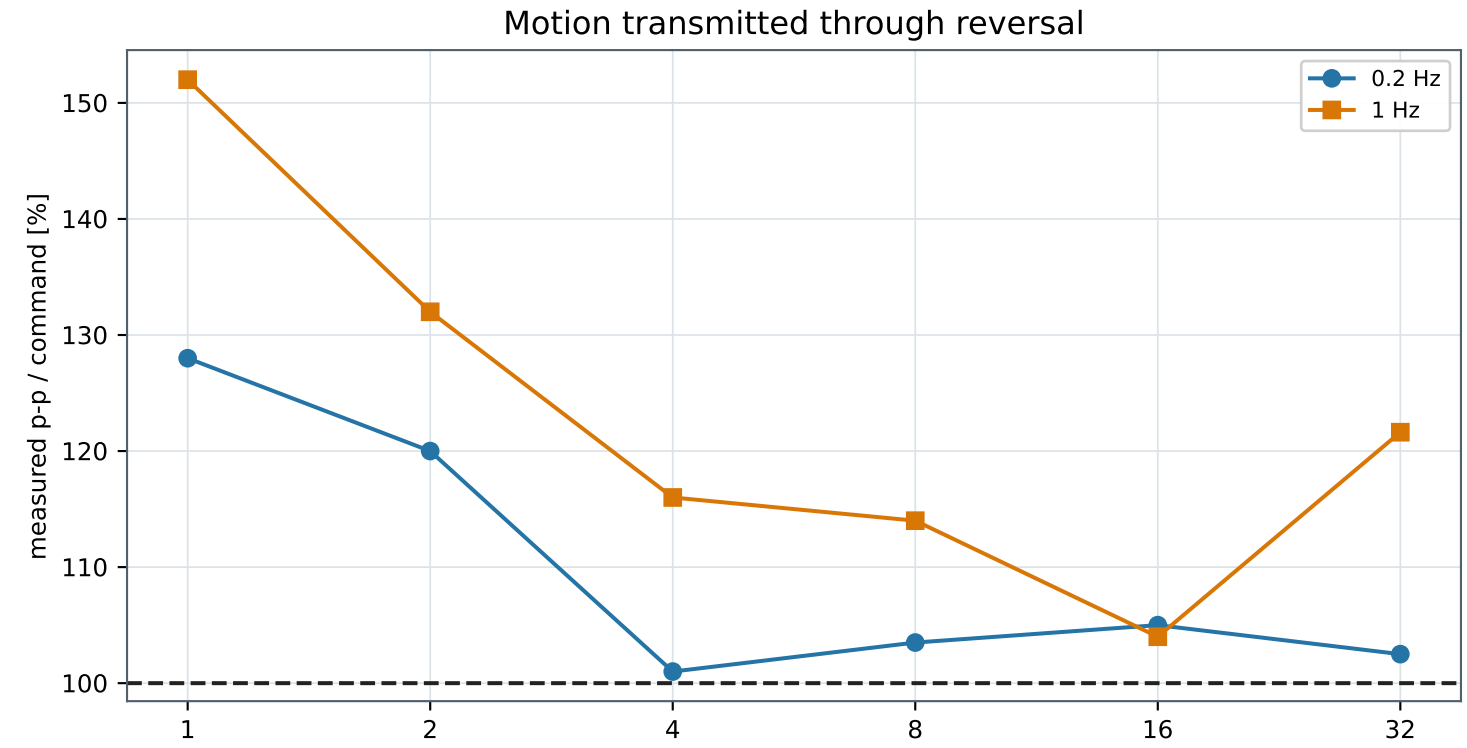


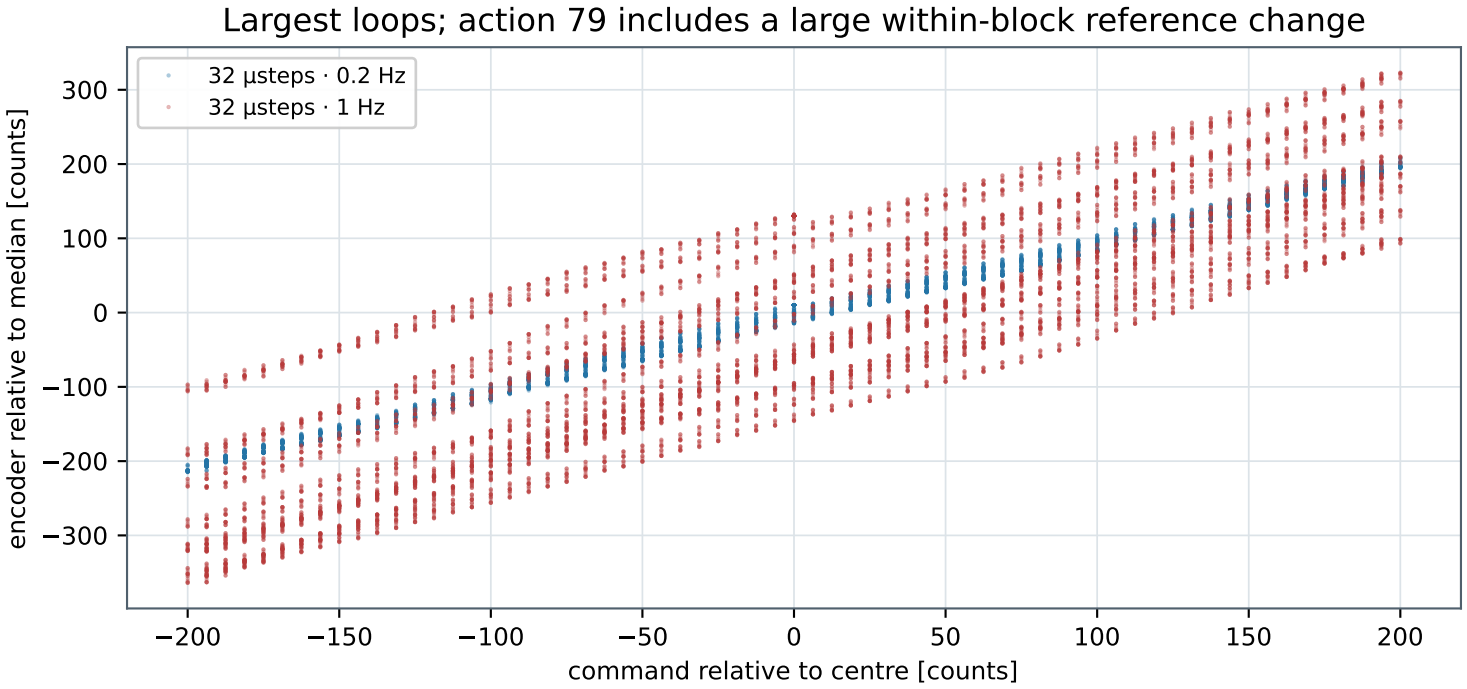


Speed gain tests count accumulation rate. A local baseline prevents earlier lost position from masquerading as velocity lag; it does not repair global reference changes.

Small triangular reversals and friction deadband

Response measured between samples at commanded positive and negative extrema.

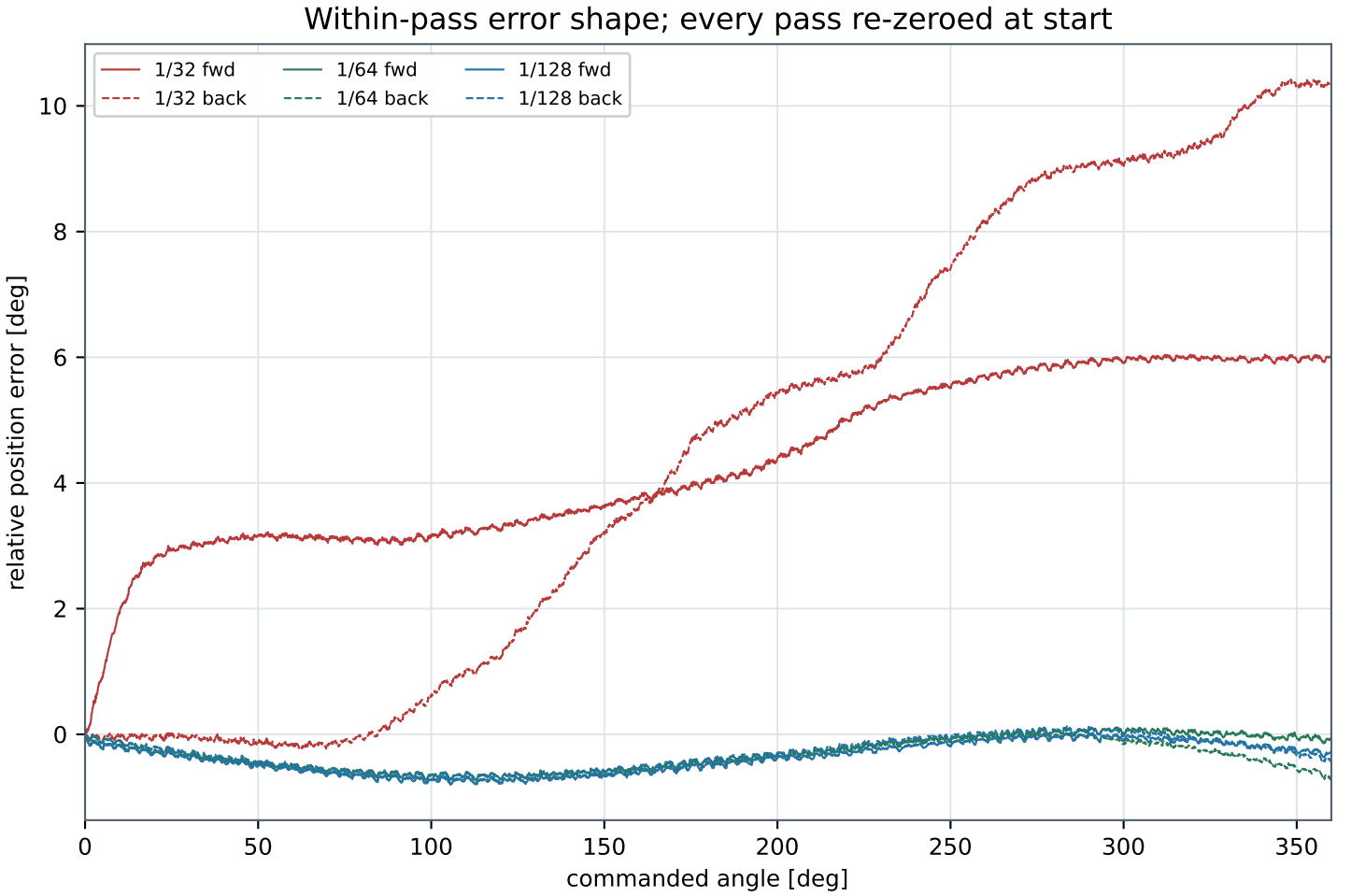




A closed narrow loop indicates repeatable following. Action 79 must not be reduced to one gain number: its +234-count boundary change is qualitatively different from ordinary hysteresis.

Fine-microstep resolution campaign

Separate one-turn forward/back tests at 1/32, 1/64, and 1/128.



1/64 AND 1/128

Mean steps 1.560 and 0.780 counts. Single-step responses 64% and 27% of ideal. Smooth 1/rev structure is consistent with eccentricity, not proof.

1/32 DATA QUALITY WARNING

The encoder accumulated +20,335 and −20,572 counts for nominal ±20,000 turns. Gradual over-count points to ringing or coupling slip. Repeat before using this trace.

One encoder count is 64.8 arcsec. An ideal 1/128 microstep is only 0.781 count, so integer quantisation dominates individual increments; averaging is essential.

Conclusions, limitations, and a defensible next calibration

Concise interpretation for laboratory discussion.

1 · WHAT THE DATA SUPPORT

The direction-paired static map resolves a 0.382°-amplitude 1/rev term plus electrical/full-step harmonics. Average 1/64 and 1/128 motion is near nominal although individual microsteps are quantised and partly effective.

2 · WHAT THE DATA DO NOT SUPPORT

V1 radial runout cannot become absolute angular calibration with present metadata. A 1/rev harmonic is not uniquely eccentricity: the motor can contain once-per-rev structure. Later blocks do not share the opening reference.

3 · RECOMMENDED CALIBRATION EXPERIMENT

Record encoder angle and dial displacement from a common trigger over several slow turns. Register index to dial high spot; record scale radius and readhead geometry; repeat after remounting. Prefer an independent angular reference.

Reproducibility. Static values use the final 28% of each dwell. Direction samples are matched at identical commands. Spectra remove offset and linear closure trend. Step responses align marked edges. Velocity fits use the central 80%. No eccentricity correction is applied to source data.

Sources: the 2026-10-13 motor-identification run, fine-microstep run, and v1 dial CSV. Machine-readable metrics and page images are stored in analysis/motor_encoder_report/.